

Nagraj Desai

AI Engineer

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SUMMARY

AI Engineer with 2+ years building production LLM systems — RAG pipelines, agentic workflows, and GPU-optimized inference (vLLM, NVIDIA Dynamo, Triton) serving 300K+ daily requests for enterprise-scale workloads.

EXPERIENCE

AI Engineer

April 2025 – Present

Micropoint Computers (NVIDIA Partner)

- Served **300K+ LLM requests/day** using NVIDIA Dynamo with KV-aware routing.
- Reduced **LLM request drop rate from 87% to 7%** for workloads with 200 concurrent users.
- Reduced **VLM E2E latency from 4 min to 1.5 min** using encoder-decoder disaggregated inference with NVIDIA Dynamo.
- Built a distributed **Facial Recognition System (FRS)** supporting ~100 crore identities using MinIO, Redis, and Milvus for scalable multi-GPU processing and vector search.
- Managed **bare-metal Kubernetes clusters** with NVIDIA H100/H200 GPUs for large-scale AI workloads, with Prometheus and Grafana monitoring.

Data Science Intern

Jul 2024 – Jan 2025

Taabi Mobility Ltd. (CEAT Tyres)

- Designed and deployed a **vehicle health and event prediction platform** using Python and GraphQL, enabling real-time monitoring and diagnostics while increasing prediction accuracy to **99.62%**.
- Enhanced the **data processing pipeline** through performance optimizations, reducing processing time by **58%** (2 hours to 50 minutes) and improving overall system throughput and reliability.

TECHNICAL SKILLS

Programming: Python, SQL

AI & Machine Learning: PyTorch, Scikit-learn, Machine Learning, Deep Learning, Natural Language Processing (NLP), Transformers, Large Language Models (LLMs), Statistics, Exploratory Data Analysis (EDA)

Generative AI: Retrieval-Augmented Generation (RAG), Agentic AI, LangChain, LangGraph, Model Context Protocol (MCP), Model Fine-Tuning (LoRA, QLoRA), Vector Databases, ASR

AI Infrastructure & MLOps: Docker, Kubernetes, vLLM, NVIDIA Dynamo, NVIDIA Triton Inference Server, NVIDIA NIM, TensorRT, AWS (EC2, IAM, S3), Linux

Data Engineering & Development: MySQL, FastAPI, REST APIs, Version Control(Git), MS Excel

KEY PROJECTS

AI Voice Agent — Stack: Python · FastAPI · NVIDIA NIM · Riva gRPC · WebSockets

- Engineered a **real-time ASR → LLM → TTS voice agent**, achieving approximately **3–4s time-to-first-audio (TTFA)** through streaming inference and audio generation.
- Reduced **perceived response latency by 60%+** using **clause-level chunking and 3 parallel TTS workers**, with ordered audio sequencing for continuous, gapless playback.
- Built **browser-side VAD and low-latency interruption handling**, enabling users to interrupt, redirect, and resume conversations during ongoing agent responses.

NexusAI: Enterprise Agentic System — Stack: NVIDIA NeMo, NVIDIA NIM, OpenAI GPT-OSS, MCP, Qdrant, RAG, Arize Phoenix, Python

- Built a enterprise-focused **agentic AI platform** using **NVIDIA NeMo Toolkit, NVIDIA NIM, and OpenAI GPT-OSS**, enabling agentic workflows with **tool calling** across a financial MCP server and live **web search**.
- Implemented a **Qdrant-backed RAG pipeline** with **reranking** to improve contextual retrieval accuracy, relevance, and response quality for enterprise use cases.
- Integrated **Arize Phoenix** for end-to-end **tracing, telemetry**, and performance monitoring, providing observability into agent workflows, retrieval quality, and model responses.

EDUCATION

Post Graduation Diploma in Data Science

2023 – 2024

IIIT Bangalore – CGPA: 3.97/4

B. Tech

2019 – 2023

Shivaji University, Kolhapur – CGPA: 8.94/10

CERTIFICATIONS

NVIDIA Technical Curriculum (DGX, AI, Compute, Generative AI, 2026)